

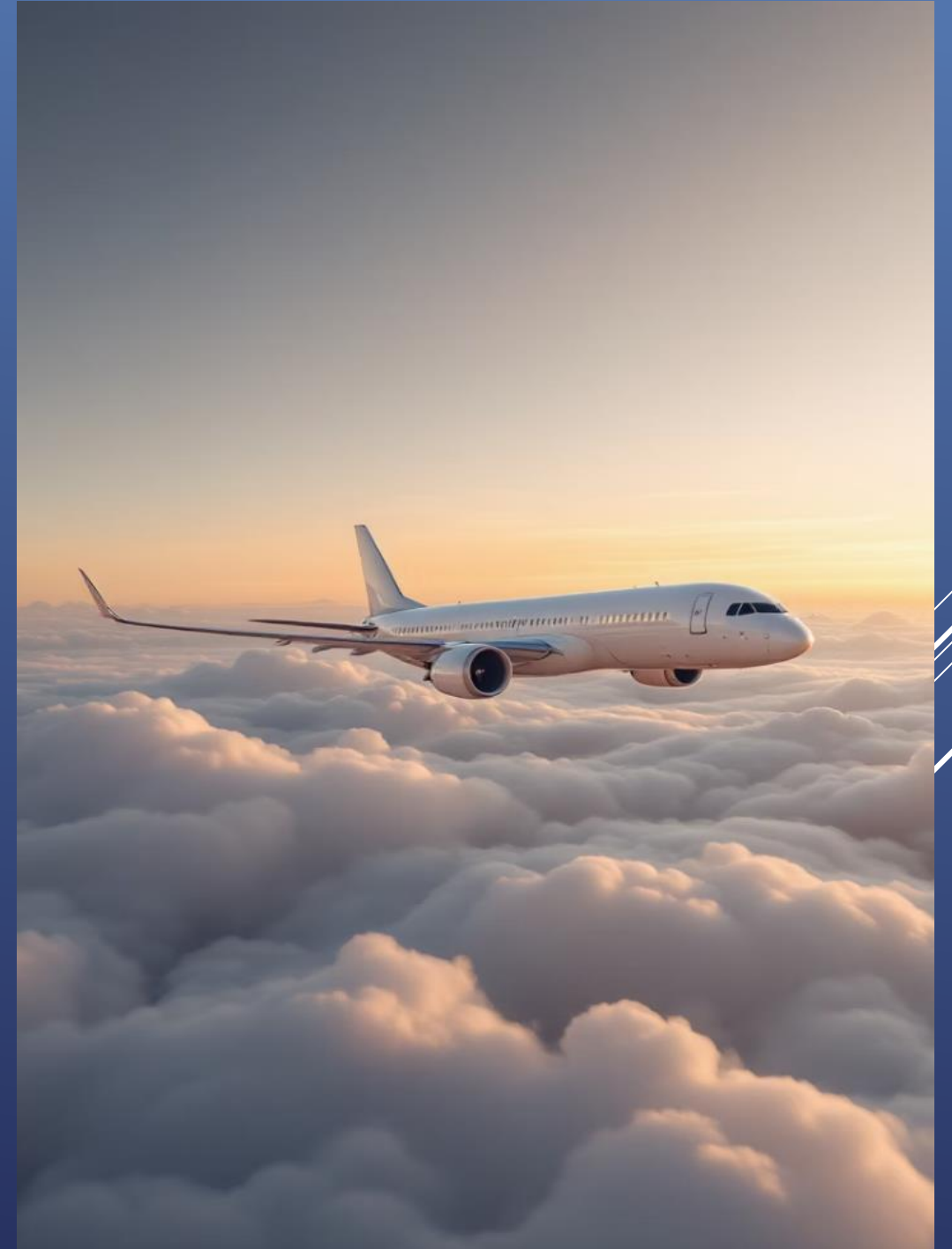
AIRLINE RESERVATION SYSTEM

MALIHA ZAFAR

MALAIKA IBRAR

Airline Reservation System

An airline reservation system (ARS) is a computerized software application that enables airlines to sell their seat inventory and manage flight bookings. It contains detailed information on flight schedules, fares, and maintains a database of passenger reservations (passenger name records) and issued tickets if applicable.



KEY FEATURES

- Flight search with multi-city and multi-stop options.
- Comprehensive flight details: fare rules, baggage info, promotions.
- Passenger seat selection (economy, business, first class).
- Secure online payment and e-ticket generation.
- Reservation management: booking, cancellation, modification.
- Dynamic seat availability and pricing.
- User roles: casual users, registered users, administrators

SQL Queries & Integration:

Cancellation	Flight search & listing
<pre>DELETE FROM bookings WHERE id = ? UPDATE flights SET \$seat_column = \$seat_column + 1 WHERE id = ?</pre>	<pre>FROM flights \$whereSql ORDER BY departure_date, departure_time SELECT * FROM flights ORDER BY departure_date, departure_time -- "All flights" table</pre>
Booking a seat	Payment
<pre>SELECT \$seat_column FROM flights WHERE id = ? FOR UPDATE INSERT INTO bookings (flight_id, passenger_name, seat_class) VALUES (?, ?, ?) UPDATE flights SET \$seat_column = \$seat_column - 1 WHERE id = ?</pre>	<pre>SELECT f.price_economy, f.price_business, f.price_first, b.seat_class FROM bookings b JOIN flights f ON b.flight_id = f.id WHERE b.id = ? INSERT INTO payments (booking_id, payment_method, card_number, pin, amount) VALUES (?, ?, ?, ?, ?)</pre>

Security Measures

- **PDO Prepared Statements:** Prevent SQL injection by separating SQL code from user input.
- **No Direct Variable Interpolation:** All user inputs are parameterized.
- **Input Validation & Sanitization:** Trimming, type checks, and escaping output to prevent XSS.
- **Transaction Management:** Ensures data integrity during critical operations.
- **Password Hashing:** Secure storage and verification of user passwords

Validation & Error Handling

Input Validation:

- Trimming inputs, required field checks.
- Type and value validation (e.g., seat class, IDs).
- Uniqueness and existence checks (e.g., user signup, booking).

Error Handling:

- All database operations in try/catch blocks.
- User-friendly error messages.
- Output escaping to prevent XSS



Airline Reservation System

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Search Flights

Departure

e.g. Pakistan

Destination

e.g. United States

Date

mm/dd/yyyy



Search

Available Flights

#	Flight Number	Departure	Destination	Date	Time	Economy Seats	Business Seats	First Class Seats	Prices (E/B/F)	Book
1	AI101	Delhi	Mumbai	2025-06-01	08:00	99	20	4	\$4,500.00 / \$8,000.00 / \$12,000.00	Economy Your Name Book

Show More Flights

Upgrade Booking

Booking ID

Enter your Booking ID

Upgrade to Class

Economy



Upgrade

Cancel Booking

Booking ID

Enter your Booking ID

Cancel Booking

Show More Flights

Customer Feedback & Reviews

Your Name

Rating

Select rating



Your Feedback

Submit Feedback

Recent Feedback

John Doe

Rating: ★★★★★

Excellent flight experience!

May 25, 2025, 4:11 pm

THANK YOU